

Day 8 — Tirthaji's Sūtra System in Context

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name at least 5 of Tirthaji's 16 sūtras, explain in 2–3 sentences when and by whom *Vedic Mathematics* was published, and state honestly that the “Vedic” attribution is academically contested.

Materials

- Printed handout: the list of 16 sūtras + 13 sub-sūtras (from Microsoft Word - sutras1.pdf).
- Optional: a paragraph excerpt from Tirthaji's preface (teacher's prepared synopsis).

Lesson flow

Warm-up (3 min)

- Recall: name the sūtra behind the $\times 11$ trick. (*Ekādhikena Pūrveṇa*.)
- Name the sub-sūtra behind $\times 12$ – $\times 19$. (*Anurūpyeṇa*.)

Core (25 min) — The 16 sūtras and the provenance question

- The 16 sūtras.** Hand out the list. Walk through abbreviated:

#	Sanskrit	English gloss
1	<i>Ekādhikena Pūrveṇa</i>	“By one more than the previous”
2	<i>Nikhilam Navataścaramaṁ Daśataḥ</i>	“All from 9 and the last from 10”
3	<i>Ūrdhva-tiryagbhyāṁ</i>	“Vertically and crosswise”
4	<i>Parāvartya Yojayet</i>	“Transpose and apply”
5	<i>Śūnyaṁ Sāmyasamuccaye</i>	“When the sum is the same, that sum is zero”
6	<i>(Ānurūpye) Śūnyam Anyat</i>	“If one is in ratio, the other is zero”
7	<i>Sankalana Vyavakalanābhyām</i>	“By addition and by subtraction”
8	<i>Pūraṇāpūraṇābhyām</i>	“By completing or non-completing”
9	<i>Calana-Kalanābhyām</i>	“By differentiation and integration”
10	<i>Yāvadūnam</i>	“Whatever the extent of its deficiency”
11	<i>Vyaṣṭi-Samaṣṭiḥ</i>	“Specific and general”
12	<i>Śeṣānyaṅkena Caramena</i>	“The remainders by the last digit”
13	<i>Sopāntya-dvayam Antyam</i>	“The penultimate and the last”
14	<i>Ekanyūnena Pūrveṇa</i>	“By one less than the previous”
15	<i>Guṇitasamuccayaḥ</i>	“The product of the sum is the sum of the

#	Sanskrit	English gloss
		products”
16	<i>Guṇaka Samuccayaḥ</i>	“All the multipliers”
	Plus 13 sub-sūtras (including <i>Anurūpyeṇa</i>).	

2. Who was Tirthaji?

- **Bharati Krishna Tirthaji** (1884–1960). Śaṅkarācārya of Govardhana Maṭha (Puri) — i.e. head of one of the four major Hindu monastic seats founded by Ādi Śaṅkara.
- Studied at Madras Christian College. Multi-lingual scholar.
- Wrote the manuscript of *Vedic Mathematics* sometime between roughly 1911 and 1918 according to its editor’s foreword.
- The manuscript was lost or unfinished for decades.
- Tirthaji **died in 1960**.
- The book was **published in 1965**, five years after his death, edited by Vasudev Sharan Agrawala.

3. The provenance question. “Tirthaji claimed the 16 sūtras come from a Vedic text called the Atharva-veda Paṛiśiṣṭa — a supplement to the Atharva Veda.”

- **The problem:** No other Sanskritist has produced a manuscript of this *Paṛiśiṣṭa* containing these 16 sūtras.
- **The mathematician S. G. Dani** (TIFR, IIT Bombay) wrote a careful critique in 1993, reprinted in *Resonance* (October 2001), pointing out:
 - No manuscript evidence.
 - The Sanskrit style of the sūtras is more recent than Vedic.
 - The mathematical techniques have 19th–20th century parallels.

4. Honest framing. Two things can be true at once:

1. The $\times 11$ trick (and the rest of Tirthaji’s 16-sūtra system) **works**. It is a useful, beautiful piece of mental arithmetic.
2. The trick is **not verifiably ancient Vedic mathematics**. The best current evidence is that Tirthaji synthesised it in the early 20th century, drawing on classical Indian arithmetic + his own pedagogical genius, and named the sūtras in Sanskrit.

The synthesis itself is a real Indian intellectual achievement. The “Vedic” naming is a 20th-century interpretive choice that scholars dispute.

5. What’s at stake. If the sūtras are Vedic (3000+ years old), they’re a major historical claim. If they’re 20th-century synthesis, **they’re still a real achievement** — just a different one. Tirthaji’s pedagogical genius is not diminished by being recent.

Discussion (15 min) — Take a position

Pose two questions to the class. Take a brief show-of-hands; then take 4–5 student statements (no debate, just listening).

1. “Does it matter whether the trick is ‘ancient’ or ‘20th-century’? Why or why not?”
2. “If it turns out to be 20th-century, should we still teach it as ‘Vedic mathematics’? What should we call it?”

End by noting: “At Senior band you’ll read Dani’s critique in full and form a deeper position.”

Wrap-up (2 min)

- Preview Day 9: “Tomorrow — mixed-practice day plus project work. Friday is the final assessment.”

Student-facing content

Tirthaji and the 16 sūtras

Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha, wrote *Vedic Mathematics* between roughly 1911 and 1918. The book was published in 1965, five years after his death.

Tirthaji claimed the 16 sūtras (including our *Ekādhikena Pūrveṇa*) come from a Vedic appendix-text. **No other Sanskritist has produced this manuscript.** S. G. Dani (1993) argues the sūtras are best understood as a 20th-century synthesis with Indian-mathematics roots, not a verifiably ancient Vedic teaching.

The trick works. The history is contested. Both can be true. The Indian intellectual achievement is real either way.

Homework

- Read the 16-sūtra list. Pick ONE sūtra (other than the two we’ve studied) and find what mathematical operation it’s used for. (Sūtras 2, 3, 4, 10, 14 are well-documented in Tirthaji 1965.) Write 3 sentences.

Differentiation

- **Tier up:** Read a one-paragraph excerpt from Dani (1993). What is Dani’s strongest critique? What does he NOT criticise?
- **Tier down:** Memorise the names of sūtras 1, 2, 3, and 14 (the four most commonly taught). Skip the historicity discussion details.

Teacher notes

- Some students (and their families) believe firmly that the trick is “ancient Vedic mathematics.” Honour the belief; don’t dismiss it. Frame your position as “the academic consensus is more sceptical, and you’ll read the evidence at Senior band.”
- Avoid civilisational chest-thumping (“ancient Indians knew this before everyone”). The Indian achievement here is the synthesis and clarity of Tirthaji’s pedagogical work; that’s enough to celebrate without overreach.

- Avoid the opposite error too — dismissing the trick as “just Western math with Sanskrit labels.” The pedagogical insight (the named patterns, the rhythm) is genuinely Tirthaji’s.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. — Sūtra list and preface.
- Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” Reprinted in *Resonance* (October 2001).
- *Microsoft Word - sutras1.pdf* (SutraGanita corpus) — the 16+13 sūtra list.